

```
%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;

% retrieve selected cuts used for further fitting
[GH_FFCutsFits,CutFitFileGH] =
recover_selected_cuts('GH_FFCutsFits','e:\SHARE\FE\Data\analysis\07_fit_with_multicu
t_all\sel_cuts');
```

```
fn = fieldnames(GH_FFCutsFits);
%nf = 11
%for nf = 2:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei800_off200GHminFF';
if ismember(selectCuts(1:3),{'msk','dat'})||...
    contains(selectCuts,'all_fit_par')
    %continue
    error('Can not fit field: %s',selectCuts);
end

fprintf('*****\n');
```

```
*****
```

```
fprintf('Processing cuts: %s \n', selectCuts);
```

```
Processing cuts: Ei800_off200GHminFF
```

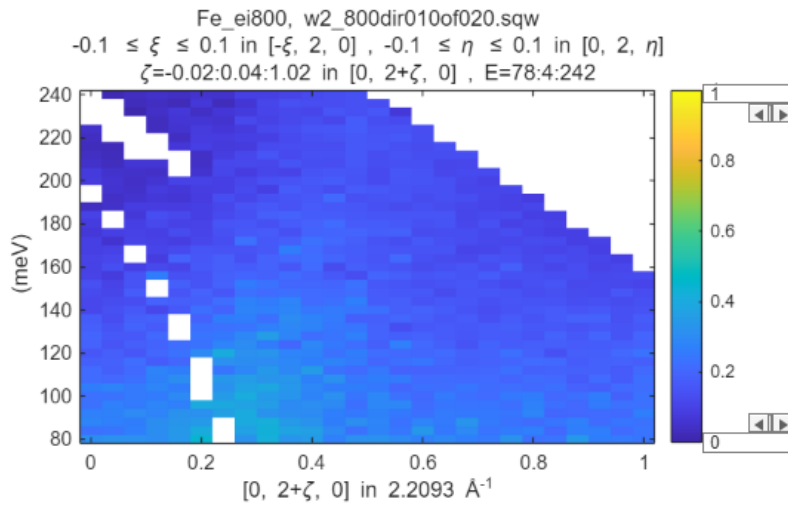
```
fprintf('*****\n');
```

```
*****
```

```
cuts_list = GH_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts,'Ei200')
    min_Ei = 30;
    mi = maps_instrument(200,600,'S');
end
if contains(selectCuts,'Ei400')
    min_Ei = 40;
    mi = maps_instrument(401,600,'S');
end
if contains(selectCuts,'Ei800')
    min_Ei = 70;
    mi = maps_instrument(786,600,'S');
end
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for j=1:numel(cuts_list)
```

```
cuts_list{j} = cuts_list{j}.set_instrument(mi);
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
```

end



```
dir_name = "GH";

dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

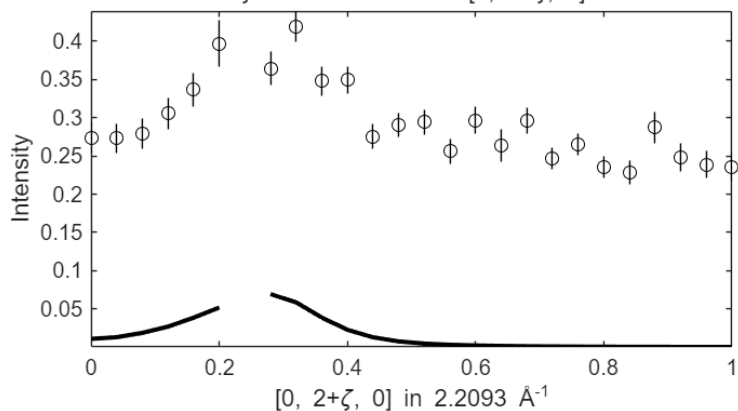
en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(min_Ei,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
GH_FFCutsFits.(selectCuts) = cuts_list;

GH_FFCutsFits = fit_multicutsFF_along_direction(...
    GH_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

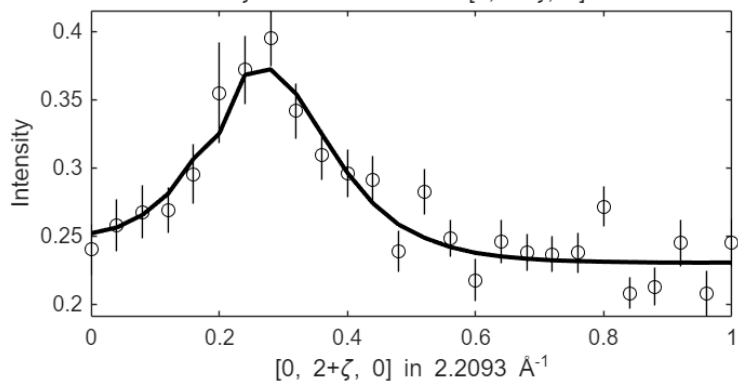
```
*****
**** En = 83±5 going to 233
*****
```

Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $78 \leq E \leq 88$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

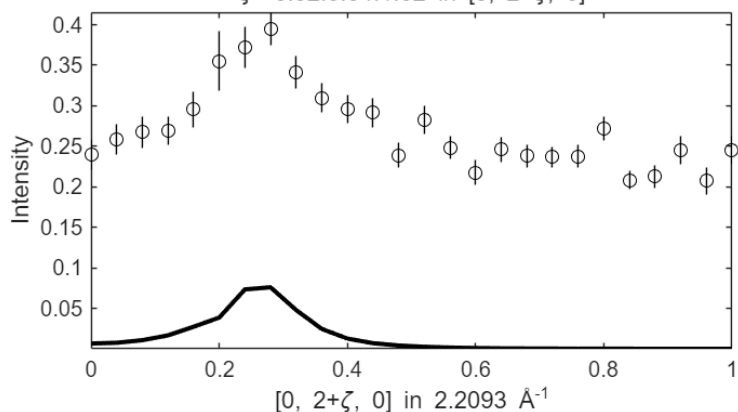


**** En = 93±5 going to 233

Fe_ei800, w2_800dir010of020.sqw
 $S=0.98 \pm 0.07$; $J_0 = 35 \pm 1.6$; $J_1 = 0 \pm 0$; $\gamma = 52.7548 \pm 4.21497$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $88 \leq E \leq 98$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

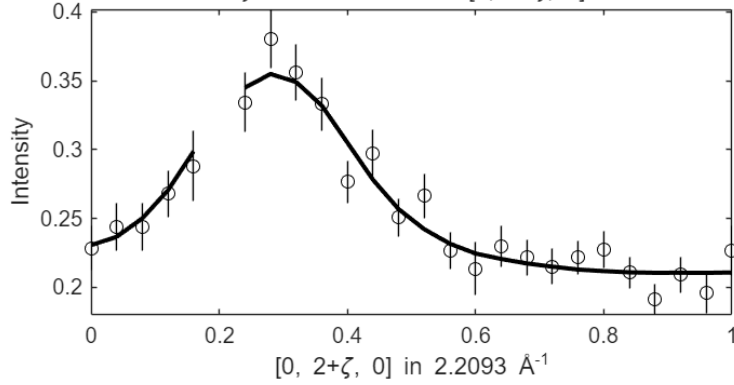


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $88 \leq E \leq 98$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

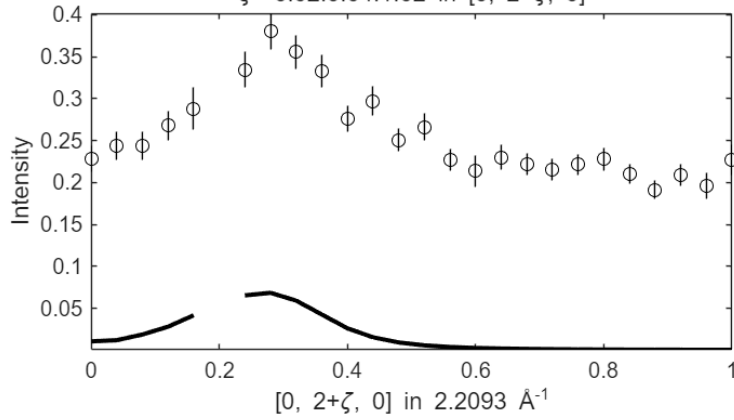


**** En = 103±5 going to 233

Fe_ei800, w2_800dir010of020.sqw
 $S = 2.3 \pm 0.12$; $J_0 = 51 \pm 1.5$; $J_1 = 0 \pm 0$; $\gamma = 181.187 \pm 8.28187$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $98 \leq E \leq 110$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

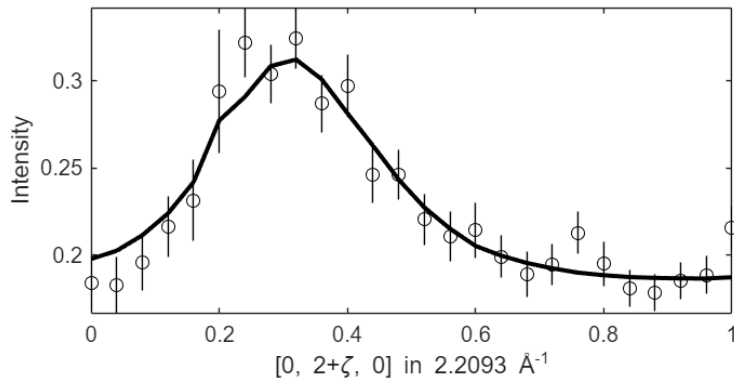


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $98 \leq E \leq 110$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

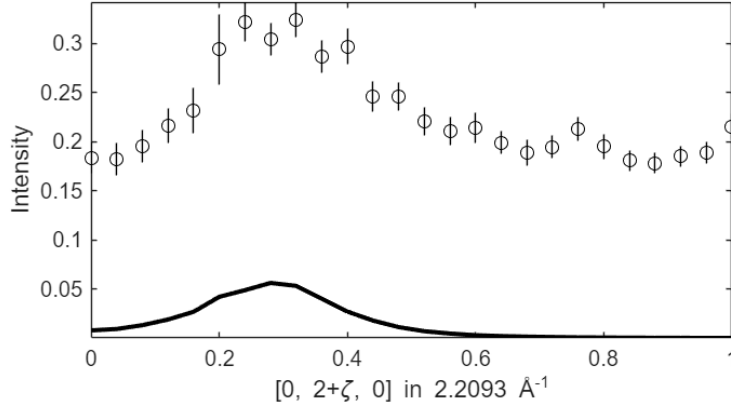


 **** En = 113 ± 5 going to 233

Fe_ei800, w2_800dir010of020.sqw
 $S = 2.1 \pm 0.11$; $J_0 = 49 \pm 1.6$; $J_1 = 0 \pm 0$; $\gamma = 192.14 \pm 8.75666$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $108 \leq E \leq 118$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

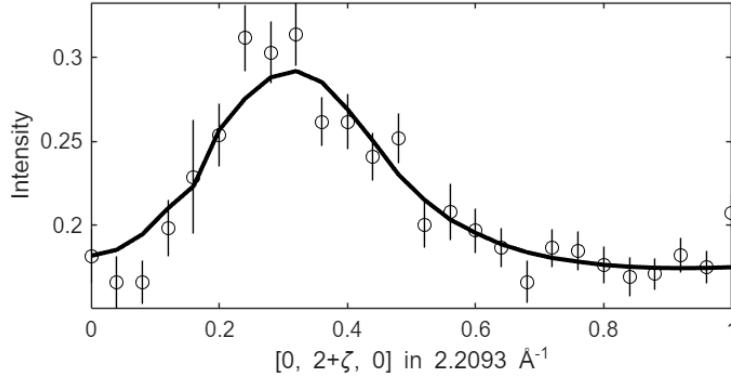


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $108 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

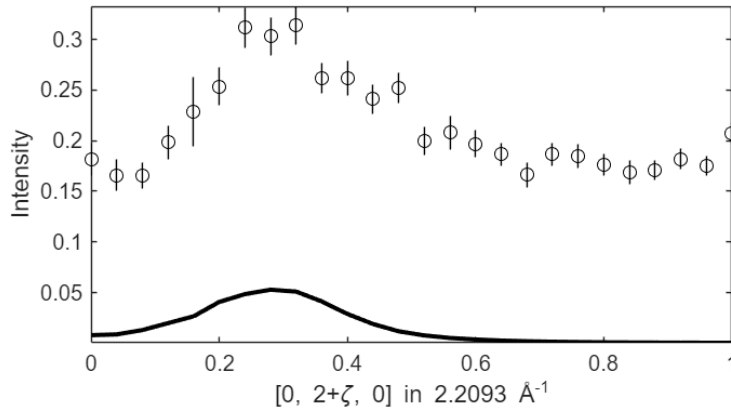


**** En = 123±5 going to 233

Fe_ei800, w2_800dir010of020.sqw
 $S = 3.1 \pm 7.4$; $J_0 = 66 \pm 1.2 \times 10^2$; $J_1 = 0 \pm 0$; $\gamma = 361.374 \pm 1393.97$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $118 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

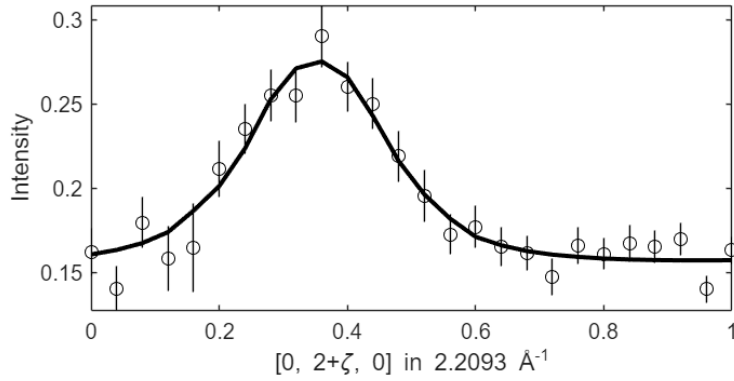


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $118 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

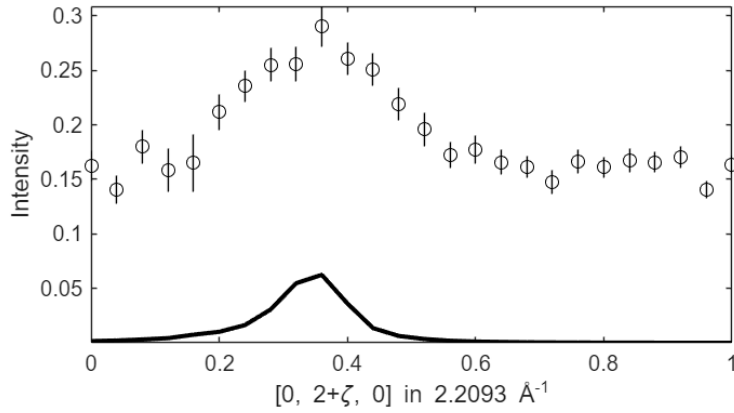


**** En = 133±5 going to 233

Fe_ei800, w2_800dir010of020.sqw
 $S=0.68\pm0.034$; $J_0= 31\pm 1.1$; $J_1= 0\pm 0$; $\gamma=35.9274\pm2.11826$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $128 \leq E \leq 1$
 $\zeta=-0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

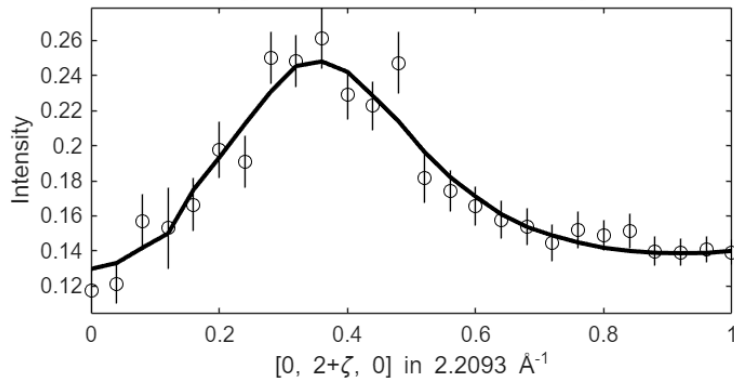


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $128 \leq E \leq 1$
 $\zeta=-0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

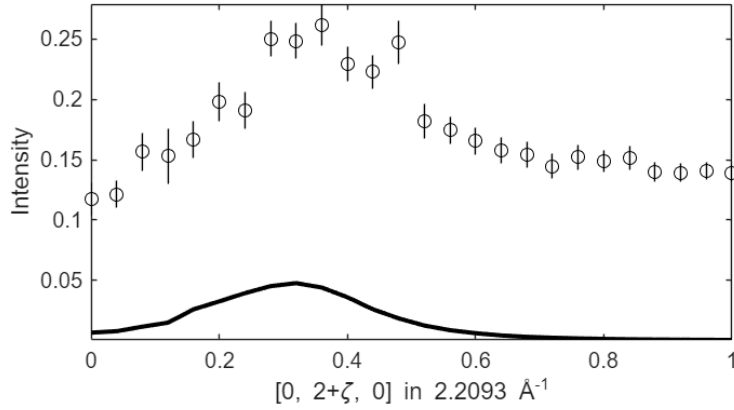


 **** En = 143 ± 5 going to 233

Fe_ei800, w2_800dir010of020.sqw
 $S= 4.9\pm 11$; $J_0= 86\pm1.6e+02$; $J_1= 0\pm 0$; $\gamma=759.701\pm2865.91$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $138 \leq E \leq 1$
 $\zeta=-0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

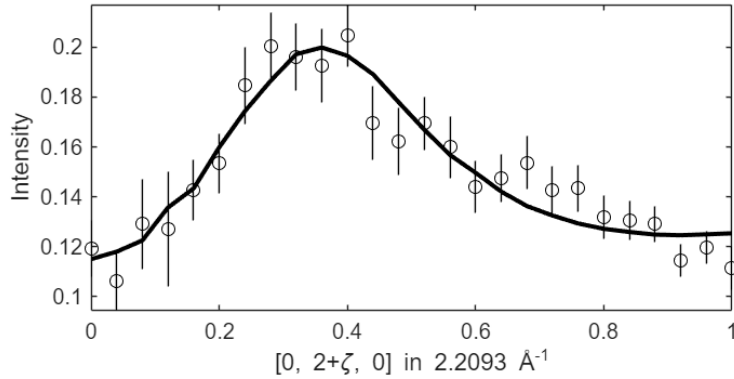


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $138 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

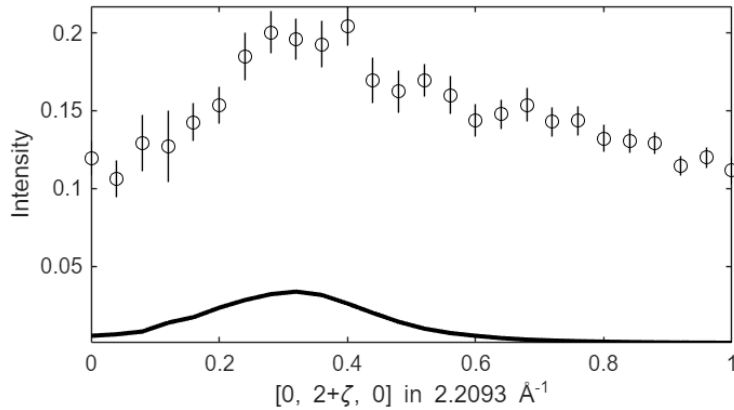


 **** En = 153±5 going to 233

Fe_ei800, w2_800dir010of020.sqw
 $S = 7.2 \pm 67$; $J_0 = 1.6e+02 \pm 1.4e+03$; $J_1 = 0 \pm 0$; $\gamma = 2778.2 \pm 48597$.
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $148 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



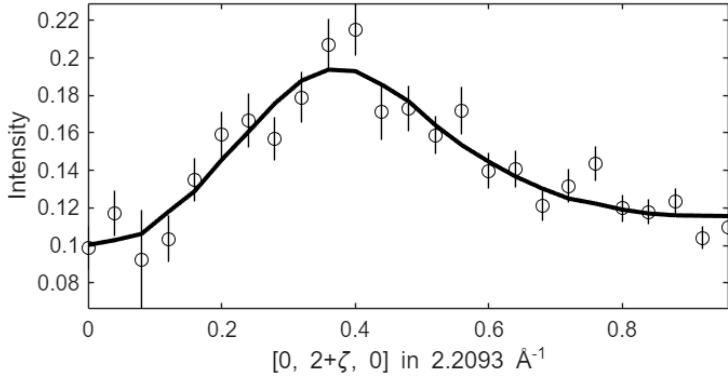
Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $148 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



 **** En = 163±5 going to 233

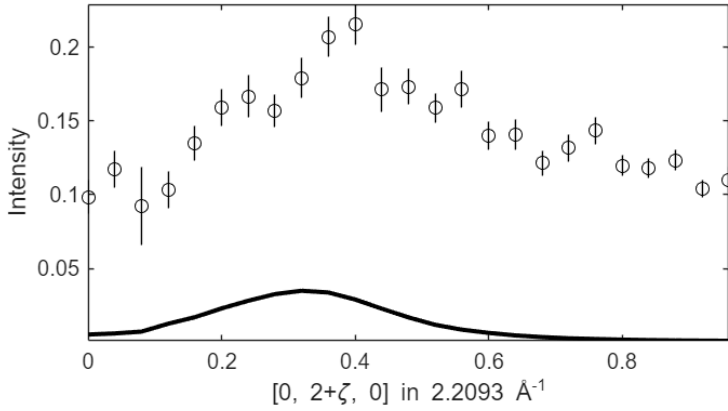
Fe_ei800, w2_800dir010of020.sqw

S= 6.3±0.54; J0=1.3e+02± 3.4; J1= 0± 0; gamma=1845.79±83.2161
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $158 \leq E \leq 1$
 $\zeta=-0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



Fe_ei800, w2_800dir010of020.sqw

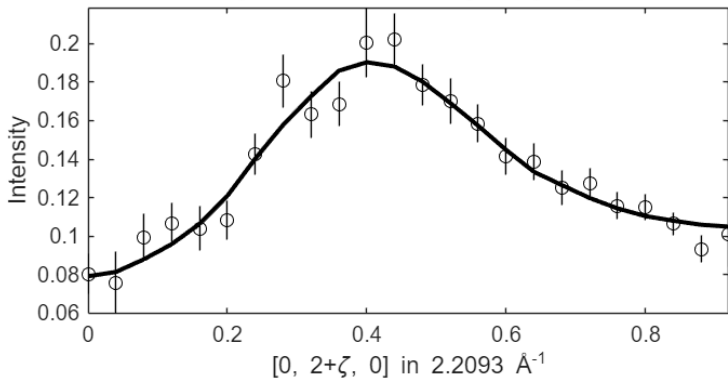
$-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $158 \leq E \leq 1$
 $\zeta=-0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



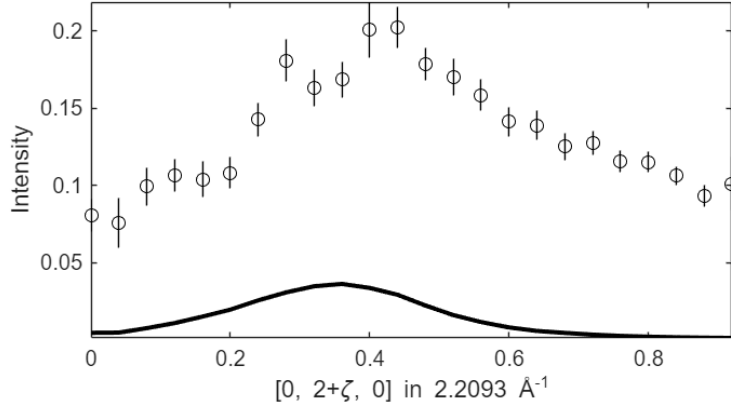
**** En = 173±5 going to 233

Fe_ei800, w2_800dir010of020.sqw

S= 3.1±0.19; J0= 59±0.71; J1= 0± 0; gamma=421.97±12.7229
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $168 \leq E \leq 1$
 $\zeta=-0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

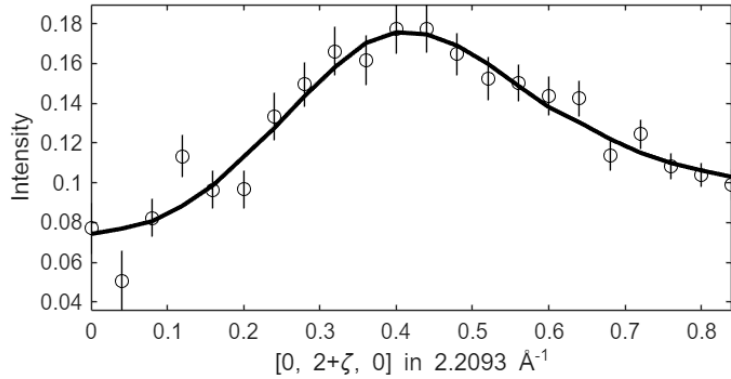


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $168 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

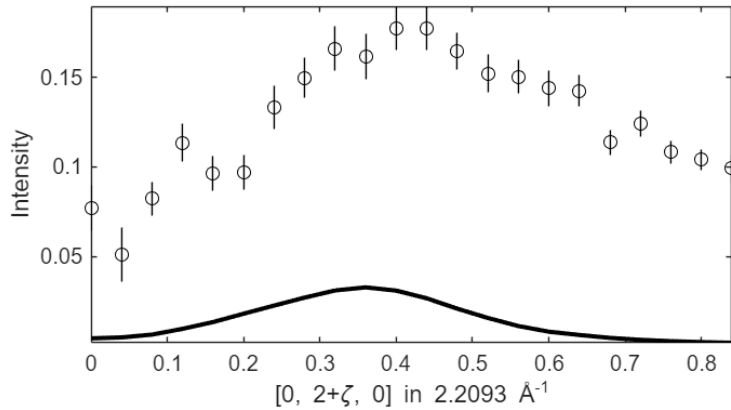


 **** En = 183±5 going to 233

Fe_ei800, w2_800dir010of020.sqw
 $S = 2.9 \pm 0.22$; $J_0 = 60 \pm 0.86$; $J_1 = 0 \pm 0$; $\gamma = 443.692 \pm 15.8678$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $178 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



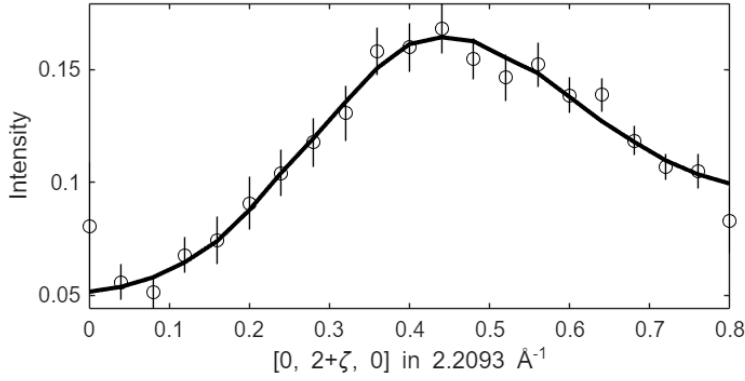
Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $178 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



 **** En = 193±5 going to 233

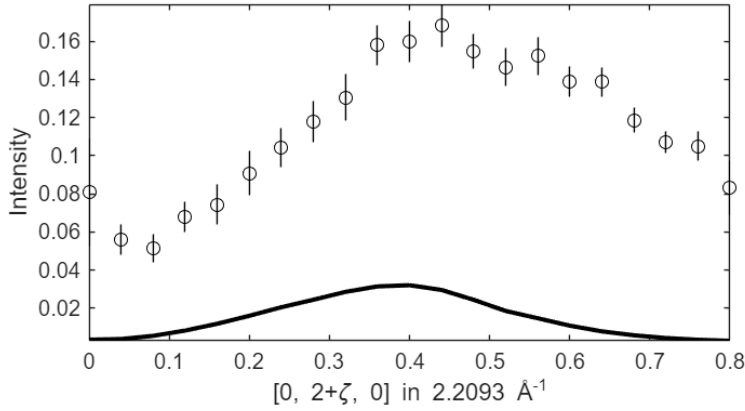
Fe_ei800, w2_800dir010of020.sqw

S= 2.7 ± 0.14 ; J0= 52 ± 0.68 ; J1= 0 ± 0 ; gamma= 363.795 ± 9.97306
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $188 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



Fe_ei800, w2_800dir010of020.sqw

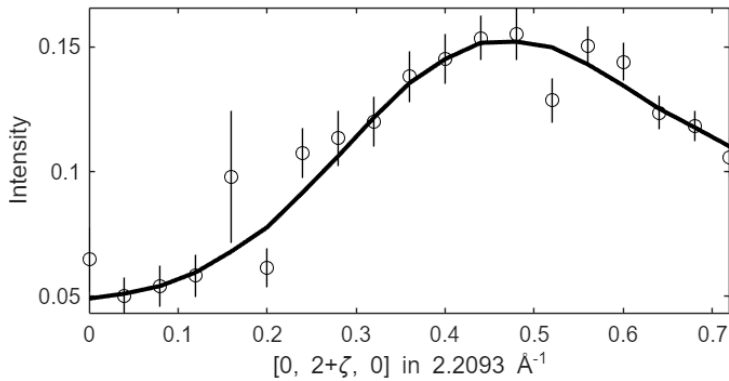
$-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $188 \leq E \leq 1$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



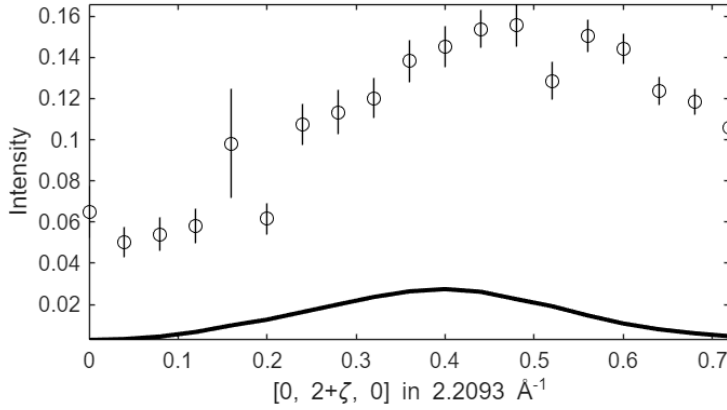
**** En = 203 ± 5 going to 233

Fe_ei800, w2_800dir010of020.sqw

S= 2.3 ± 0.2 ; J0= 48 ± 1.3 ; J1= 0 ± 0 ; gamma= 341.372 ± 15.3972
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $198 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

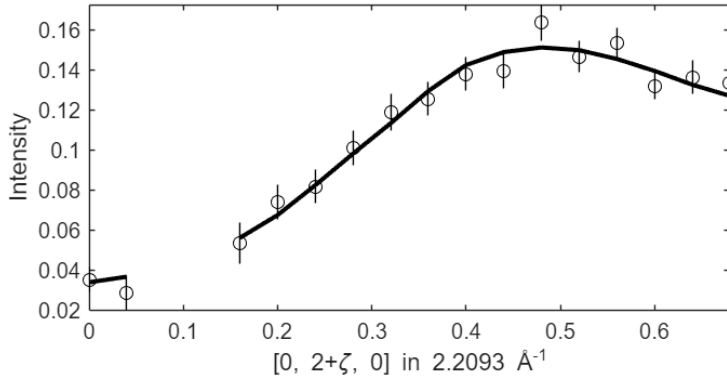


Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $198 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$

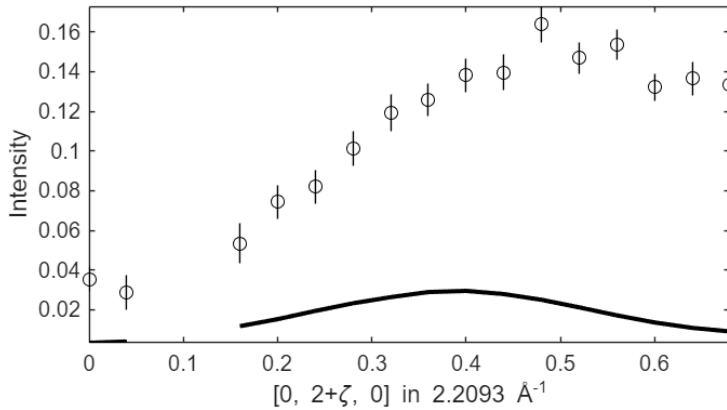


**** En = 213±5 going to 233

Fe_ei800, w2_800dir010of020.sqw
 $S = 6.2 \pm 0.52$; $J_0 = 92 \pm 4$; $J_1 = 0 \pm 0$; $\gamma = 1428.1 \pm 84.5405$
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $208 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



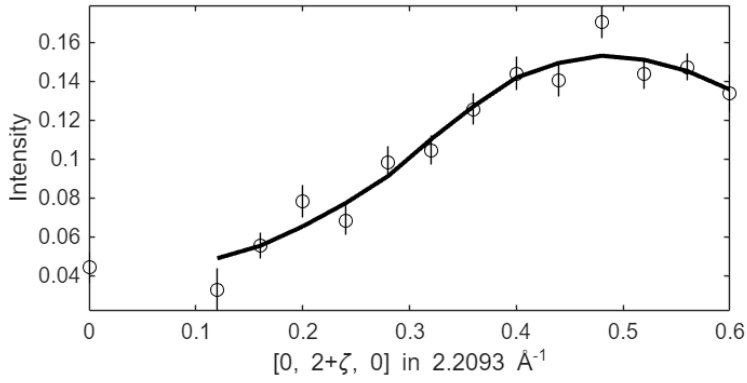
Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $208 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



**** En = 223±5 going to 233

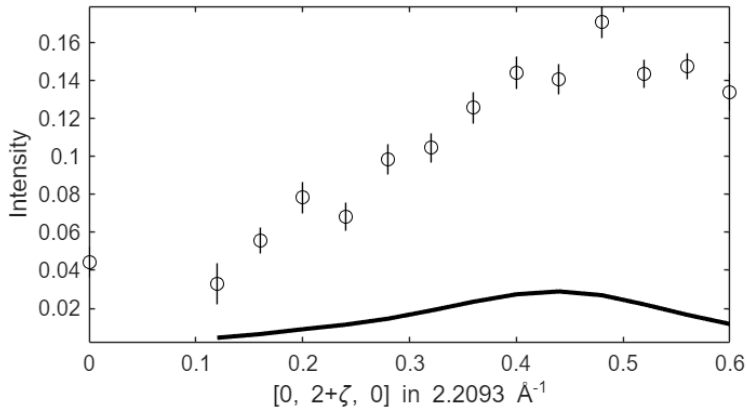
Fe_ei800, w2_800dir010of020.sqw

S= 1.4 ± 0.15 ; J0= 36 ± 1.5 ; J1= 0 ± 0 ; gamma= 155.088 ± 9.13103
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $218 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



Fe_ei800, w2_800dir010of020.sqw

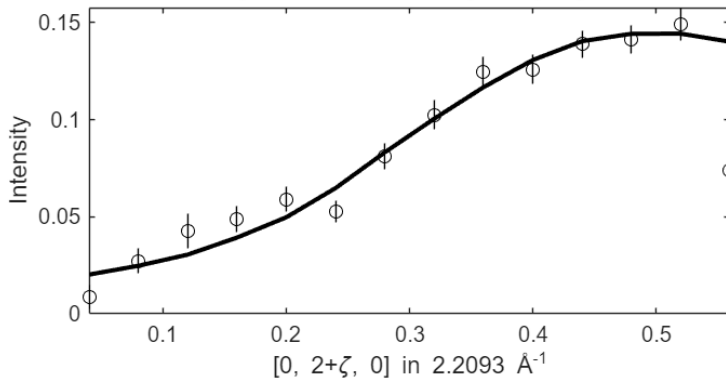
S= 1.4 ± 0.15 ; J0= 36 ± 1.5 ; J1= 0 ± 0 ; gamma= 155.088 ± 9.13103
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $218 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



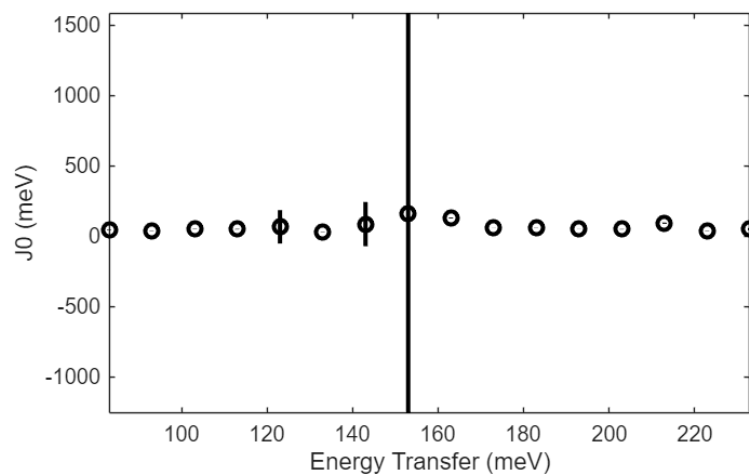
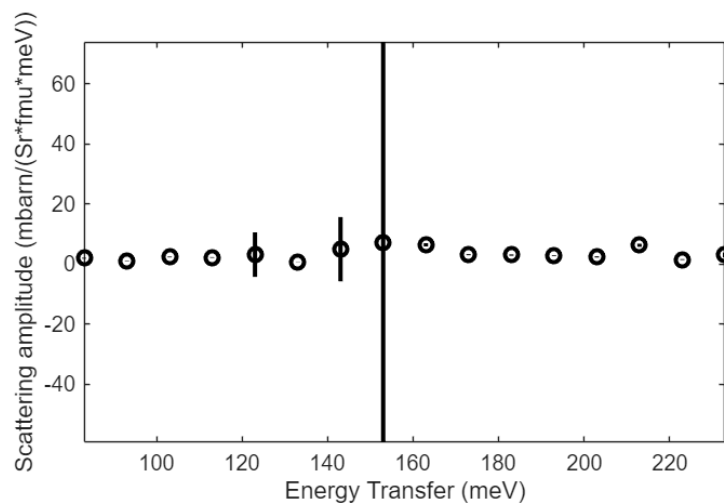
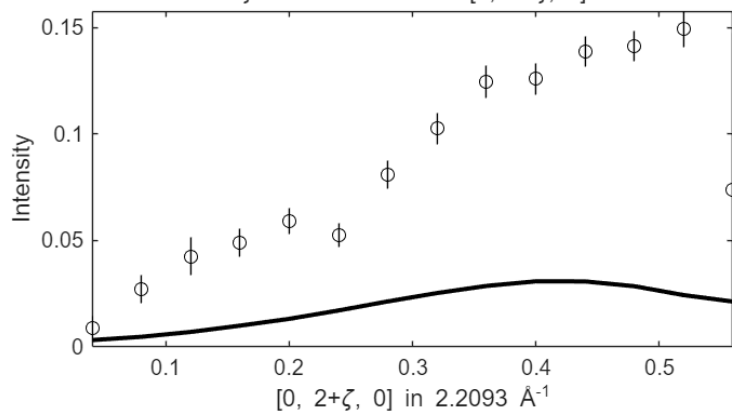
**** En = 233 ± 5 going to 233

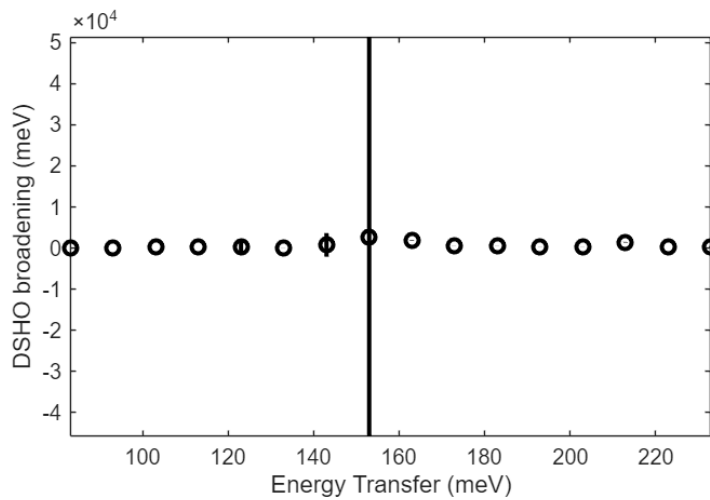
Fe_ei800, w2_800dir010of020.sqw

S= 3 ± 0.77 ; J0= 50 ± 7.5 ; J1= 0 ± 0 ; gamma= 365.243 ± 102.158
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $228 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$



Fe_ei800, w2_800dir010of020.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[-\xi, 2, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0, 2, \eta]$, $228 \leq E \leq 2$
 $\zeta = -0.02:0.04:1.02$ in $[0, 2+\zeta, 0]$





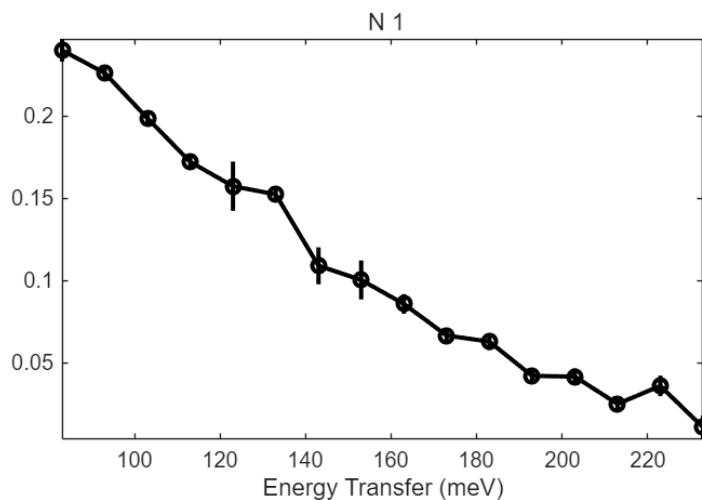
```

c1 = GH_FFCutsFits.(selectCuts);
GH_FFCutsFits.(selectCuts) = [c1{:}];

beep
beep
beep

all_fit_par = GH_FFCutsFits.([selectCuts, '_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;

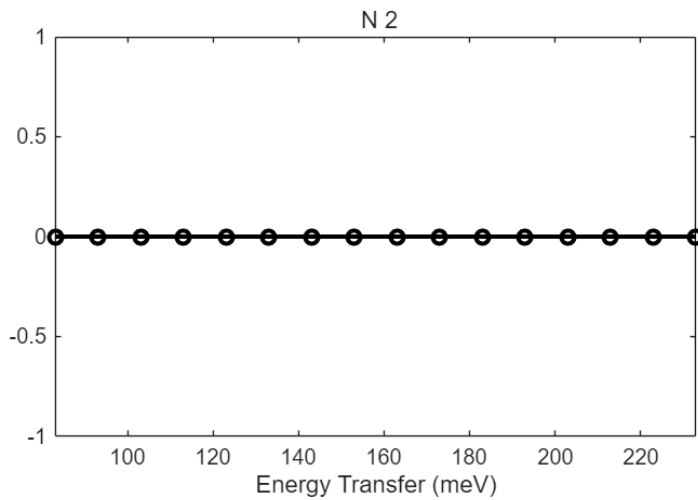
```



```

pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure

```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fig
ure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fig
ure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_GH.m';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_GH_',selectCuts,'.pdf']);
export(thisFile,targ_file);
```

```
fit_src_struct = GH_FFCutsFits;
save(CutFitFileGH,"fit_src_struct",-v7.3');
close all;
%end
```